

Protocol: Ethnicity-based System Dynamics Model of Measles Vaccinations in Birmingham and Solihull

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1 Background

Measles is a highly infectious viral disease that can cause serious complications and death [1]. In high-income countries, complications occur in around 10-20% of cases, and the case-fatality ratio is around 0.05% [2, 3]. Clinically vulnerable groups, such as immunosuppressed individuals, infants under one year, and pregnant people, are more likely to develop complications [4].

The first measles vaccine was licensed in the US in 1963 [5]. The measles vaccine was first introduced in the UK in 1968 and later replaced by the combined measles, mumps, and rubella (MMR) vaccine in 1988 [6]. As of 1 January 2026, varicella has been added to make the MMRV vaccine¹. In the UK, all children are offered a first dose of MMR when they turn 12 months and a second dose at 18 months [7]. The World Health Organisation (WHO) estimates that between 2000 and 2024, the measles vaccine has prevented 59 million deaths [8].

The UK achieved measles elimination in 2016 [7]. However, measles outbreaks are becoming increasingly common in the UK and globally due to falling vaccine coverage [9]. The top half of Figure 1 shows that there has been a series of measles outbreaks in England since January 2012.

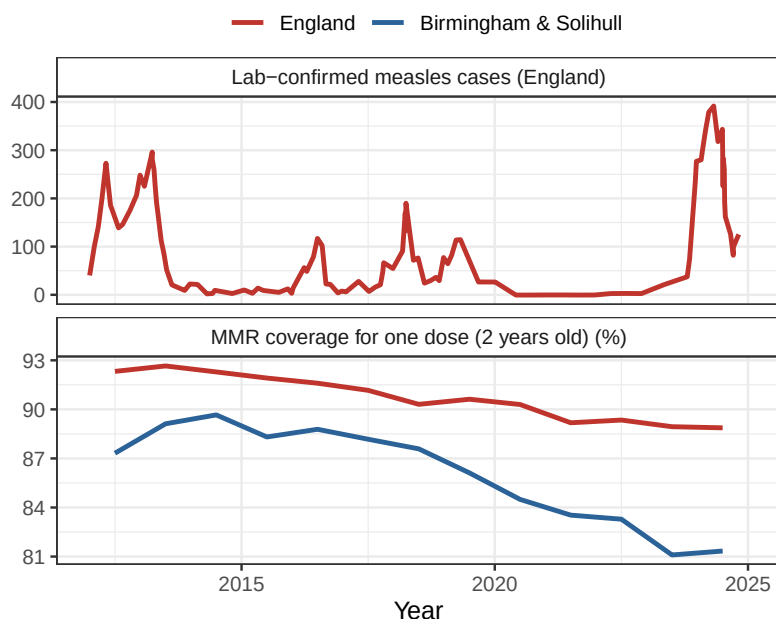


Figure 1: (Top) Monthly lab-confirmed cases of measles in England [10]. (Bottom) Yearly MMR coverage for one dose at two years old [11].

The bottom half of Figure 1 shows these outbreaks occurring alongside falling national vaccination coverage from a peak of 92.7% in 2013 down to 88.9% in 2024. However, in Birmingham and Solihull

¹For brevity, MMR and MMRV will simply be referred to as MMR.

(BSol), the coverage started lower and has fallen further than the England average from a peak of 89.7% in 2014 down to 81.3% in 2024 [11]. This has contributed to the region suffering large measles outbreaks [12].

Part of the reason for the falling MMR coverage is the growing scepticism in vaccines [13, 14]. While this is a trend for most vaccines, there has been particular public concern about the MMR vaccine due to the infamous, since retracted, claim by one research group that the MMR vaccine might cause autism [15–17]. However, there are driving vaccine hesitancy that vary across population groups, including misinformation, health literacy, and lack of trust in the healthcare system [18]. Some groups, such as Gypsy, Roma, and Traveller populations, also face additional barriers to accessing the vaccine [19]. As a result, significant ethnic and socioeconomic inequality has been observed in MMR coverage [20, 21].

During the outbreak in Birmingham from 13 October 2023, the measles infection rate was 3.4 times higher for those living in the most deprived quintile compared to those living in the third most deprived quintile. No infections occurred in the least deprived quintile. Additionally, the infection rate for Black African residents was eight times greater than the White British population [22]. Figure 2 shows that, except for those of Indian and Chinese heritage, children of all ethnic groups have lower MMR coverage at one dose than the White British cohort. The Black African and Black Caribbean have particularly low coverage at 74.2% and 64.0% respectively compared to 92.9% for White British [23].

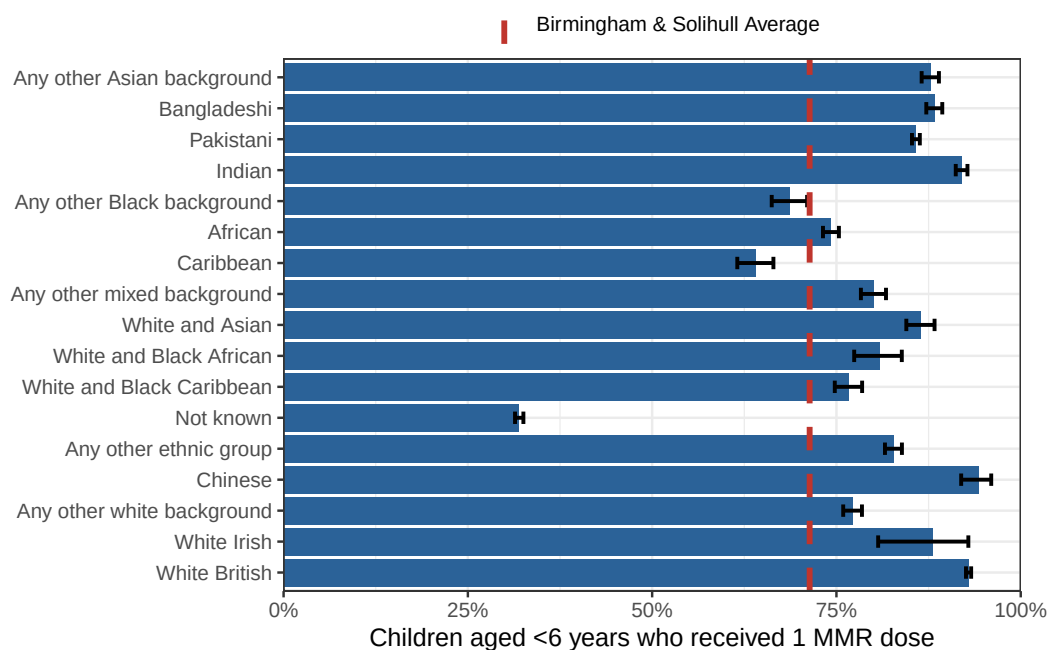


Figure 2: Percentage of children aged less than six years old living in BSol who have received at least one dose of MMR or MMRV as of 04/02/2026 [23]. The error bars indicate the 95% CI calculated using the Wilson method.

Among the Core Cities of England and Wales², Birmingham has the highest level of ethnic diversity. Data from the 2021 Census show that just under half (48.5%) of BSol residents identified as White British, meaning that a majority (51.5%) reported a different ethnic background. This is almost double the England average of 26.5%. The largest ethnic minority groups were Pakistani (14.9%), Indian (5.7%), Black African (5.0%), and Other White (3.8%) [24].

All six WHO regions have declared measles elimination goals [1]. The UK Measles and Rubella elimination strategy 2019 set the MMR vaccination target, based on WHO recommendation, to 95% of two doses by 5 years of age in order to achieve and maintain measles elimination [25]. The Public Health England (PHE) immunisation inequalities local action plan template outlines that factors

²England and Wales Core Cities include Birmingham, Bristol, Cardiff, Leeds, Liverpool, Manchester, Newcastle, Nottingham, and Sheffield.

responsible for inequitable vaccine uptake are context-specific [26]. Therefore, in order to reverse the falling measles vaccine coverage and growing health inequalities, it is essential to understand the range of factors influencing parents' decisions to vaccinate their children and how these factors are influenced by ethnicity.

This project aims to build a decision-support system in collaboration with a range of local health protection experts working in BSol. An interactive workshop will be hosted to build a conceptual model of the local system, to identify behavioural and epidemiological factors that facilitate and block flows in the system, and to identify potential interventions to improve these flows. This model, including selected interventions, will be built using a system dynamics approach. A full timeline for the project is presented in Appendix C.

2 Literature Review

This project requires a two-part literature review. Firstly, a surface-level review was conducted into the primary computational methods that have been used to simulate epidemics, including vaccinations. This review assessed the strengths and weaknesses of each method and the extent to which they have been applied to address inequalities in measles vaccination. A second review was conducted on the factors that influence parents' decisions to vaccinate their children. These reviews will be compiled into an evidence pack that will be provided to stakeholders to support the participatory modelling workshop.

A semi-structured approach was adopted to finding literature sourcing literature from PubMed and Proquest and supplemented with additional publications and gray literature found through other means. Only English language publications were considered. For simulation methods, publications were limited to those released within the last 5 years, and within the last 10 years for vaccination barriers.

2.1 Modelling Measles Epidemics

The following literature search string was used to identify methods-focused papers on the simulation of measles including vaccination:

```
("measles")
AND ("mathematical model*" OR "transmission model*" OR
     "compartmental model*" OR "SEIR" OR "System dynamics")
AND (vaccin*)
AND ("model structure" OR parameter* OR calibration OR validation)
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There were 26 and 48 papers identified by PubMed and ProQuest, respectively. After removing duplicates and screening the titles and abstracts for relevance, there were 26 papers remaining. Due to their relevance, two papers outside the given scope were also included. The papers included are presented in Table A1.

2.1.1 SEIR Models

As shown in Table A1, the majority of the identified measles simulations used a susceptible, exposed, infectious, recovered (SEIR) model or some variant thereof [27–49]. These models compartmentalise a population into distinct states with defined rates of transition from one state to another as outlined in a simple example in Figure 3. The dynamics of this model can be described by a series of ordinary differential equations (ODEs). The ODEs for the system shown in Figure 3 are outlined in Appendix A including an explanation of basic approaches to solving the equations computationally to calculate the change in population in each state over time. Model parameters such as the infection rate β , transition rate σ , and recovery rate γ , can be drawn from existing literature or estimated by fitting the model to historic data as done by Liu et al. and Chukwu with strong agreement to the historical data [28, 30].

Most of the SEIR-type simulations identified include at least one vaccinated state. Doing so enables future measles infection rates to be estimated for different vaccination scenarios. For example,

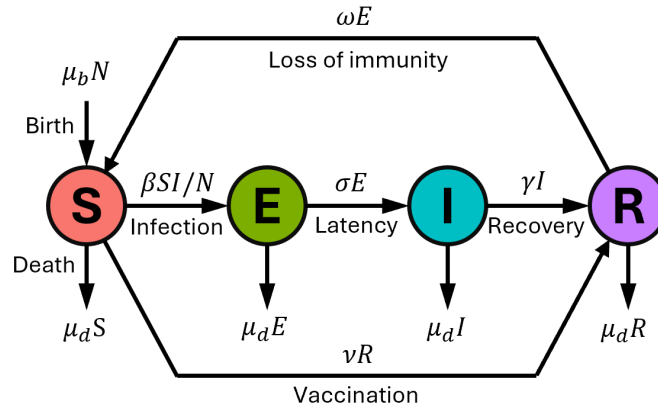


Figure 3: Example of a basic susceptible, exposed, infectious, recovered (SEIR) model, adapted from [50]. The ODEs for the system are outlined in Appendix A.

Sinigirira et al. considers the impact of varying the proportion of people being vaccinated as part of both the routine vaccination programme and an additional vaccination campaign on the number of measles infections in Nigeria [31].

One weakness of SEIR-type simulations is the inherent assumption of an idealised susceptible population with perfect population mixing [50]. This can be partially overcome by introducing additional states so that different parts of the population can be modelled with different behaviour and parameters. For example, influenza has the biggest health impact on older adults, while younger people play a more significant role in spreading the infection [51]. These age groups, therefore, need to be modelled separately as done by Waterlow et al. [52]. While measles affects people of all ages, it is most dangerous for small children [53]. Therefore, SEIR models of measles often do not model age groups separately. However, as outlined in Section 1, infection rates can vary significantly across social deprivation. For that reason, Menkir, Jbaily, and Verguet built a transmission model stratified by income [49]. This enabled the equity of different vaccination coverage strategies to be studied. One of the most challenging aspects of modelling different population groups is estimating the level of epidemiological mixing between each group.

2.1.2 System Dynamics

Two of the identified papers published in the last five years used a system dynamics approach [54, 55] as well as an additional paper outside the given publication window [56].

Like standard SEIR models, system dynamics modelling is used to model the behaviour of complex non-linear systems. In system dynamics terminology, these systems are represented using stocks, flows, and feedback loops [57]. The primary difference between these approaches is in the modelling philosophy. While SEIR systems are used to describe disease transmission dynamics, system dynamics is used to understand how the interactions between different system components generate behaviour over time. As such, system dynamics modelling often includes more abstract and difficult-to-quantify factors, including information flows and decision processes.

System dynamics has often been used in the planning of primary and secondary healthcare systems but has typically been used less often for modelling infectious disease transmission [58]. A minimal system dynamics model for infectious disease transmission modified from Ackere and Schulz is presented in Figure 4 [56].

The earliest identified paper on simulating measles vaccinations and infections using system dynamics by Ackere and Schulz present a “stylised” model of the measles system, including epidemic/medical and behavioural factors. This model was developed to capture the key dynamics of the system and to facilitate a conceptual understanding of the system and the interactions between different factors, rather than to produce accurate epidemiological predictions of future infection numbers [56]. Much like the SEIR models discussed, Ackere and Schulz segment the population into those who are susceptible, infected, recovered and vaccinated, as well as those who experienced

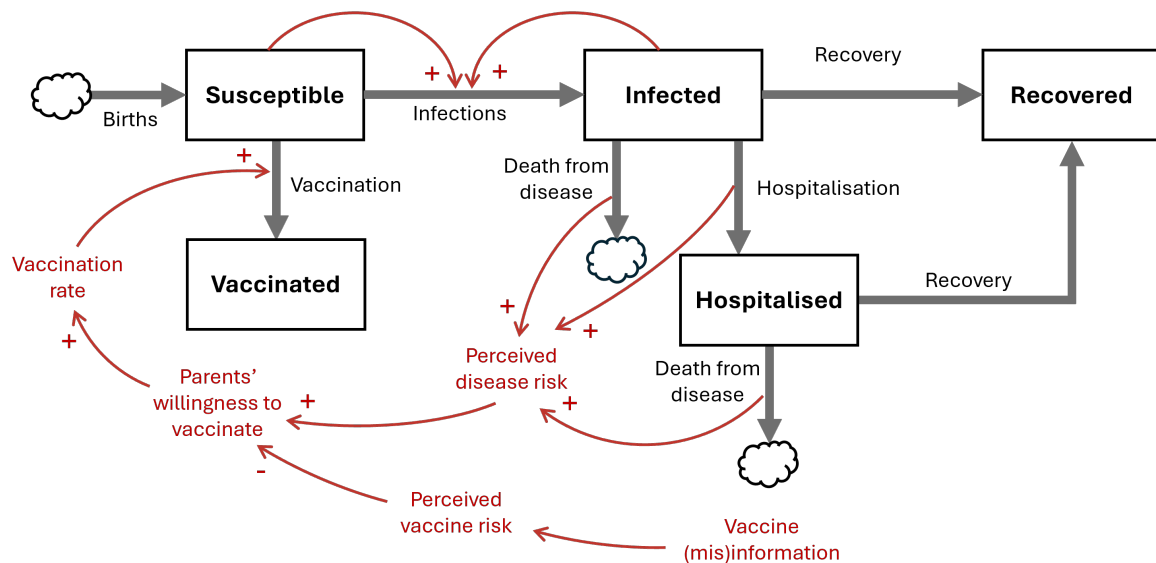


Figure 4: Minimal system dynamics model for infectious disease transmission modified from [56]. The boxes indicate the stocks (segments of the population), connected by flows in solid grey. The red lines indicate the flow of information. The clouds indicate points where the population enters (births) or leaves (deaths) the system being modelled.

severe side effects of vaccination and those who recovered from measles but with severe long-term impacts. These population states are connected by epidemiological factors such as infection rate and recovery rate, and behavioural science factors such as perceived risk of the disease and, conversely, the perceived risk of the vaccine. Parents' willingness to vaccinate is therefore a function of the number of people infected, the number vaccinated with side effects, and the impact of external information [56]. Ackere and Schulz simulate the long-term (80 years) impact of a series of 10 scenarios relating to different vaccination campaigns and minor and major vaccine scares like the 1998 Wakefield scandal [16].

Although not directly referencing Ackere and Schulz, Decouttere et al. build on this work to include additional factors, including outreach campaigns, community health workers, and physical access to vaccination, tailored specifically to the context of Rwanda [55]. However, this was only a conceptual model and not developed into a computational simulation.

Zadeh et al., in the most recent of the identified papers, attempt to create a much more complete disease transmission model, including the time from exposure to being infectious, the quarantine of exposed and infected individuals, and international travel [54]. While the authors developed this model to be applicable to any infectious disease, they demonstrated the capabilities of the model for the test case of measles, fitting the model to data from the WHO and Centres for Disease Control to predict the impact of a series of different interventions.

The focus of this project is to model the ethnicity-dependent interactions between different epidemiological and behavioural factors on the uptake of the MMR vaccine and the implications on measles outcomes. Therefore, due to its flexibility and ability to study behavioural factors, a system dynamics approach will be used. To the author's knowledge, there has not been an attempt to simulate a measles vaccination system that considers the interaction between epidemiological and behavioural factors, including the variation in these factors across ethnicity.

2.1.3 Other Methods

System dynamics and SEIR models are deterministic - given the same input parameters, the outputs will be the same. In contrast, Anwar et al. implemented a stochastic (probabilistic) model. The resulting data were used to train a neural network to approximate the dynamics of the system [59]. Such networks are typically black-box models, making the internal relationships between variables difficult to interpret, thus reducing the conceptual transparency and therefore its usefulness as a

decision support tool [60].

The methods considered above all model non-linear behaviour. However, it is also possible to develop linear models. Rodríguez-Pereira, Ramalhinho, and Sarrà, present a such a model to optimise the location of vaccination sites, the schedules of medical teams, and the routes taken between sites [61]. The authors implement a heuristic algorithm to reduce the computational cost of performing the optimisation. Due to its focus on operational systems, this approach is not appropriate for the current project.

Other important modelling approaches used to simulate the transmission of infectious disease, although not identified in this search, include agent-based models [62] and network models [63]. These approaches focus on individual interactions and contact networks, whereas the present study focuses on population-level vaccination dynamics.

2.2 Barriers to Vaccination

The following literature search string was used to identify publications on barriers experienced by parents or carers to have their child vaccinated with MMR:

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(vaccine OR vaccination OR immunisation)
AND (barrier OR hesitancy)
AND (child OR parent)
AND (measles OR MMR)
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There were 16 and 35 publications identified by PubMed and ProQuest, respectively. Five additional publications were included, four of which were grey literature based on local community engagement workshops [64–67]. The publications included are presented in Table A2. Due to their increased relevance to the present work, the following review pays particular focus on the local evidence.

A 2024 meta-analysis estimated the global prevalence of caregiver vaccine hesitancy to be 32% but only 2% for England [68]. This England estimate, based on a small sample taken in 2015, does not seem consistent with the current level of MMR coverage [69]. Another 2017 systematic review of socioeconomic differences in childhood vaccinations found MMR uptake in England to be lower amongst more affluent parents [70]. Again, the most recent study this finding was based on was published in 2011. In contrast, ONS data shows that MMR uptake is lowest in the most deprived local authorities [11].

Community engagement workshops conducted in Birmingham and Solihull, targeted at low-uptake ethnic groups, found similar overarching themes [64, 66, 67]. Language barriers and low reading ability arose as a key issue. Many parents in BSol speak little English and therefore struggle to benefit from English-only flyers while others are unable to read in any language. This highlights the importance of using multiple media forms, including video, for outreach [64, 66]. Participants also reported that the materials used to promote vaccination lacked diversity and did not represent the communities they were from [66]. This finding mirrors that of a 2024 systematic review on the barriers and enablers to MMR uptake among disadvantaged, minority, or underserved populations [71].

Concerns about the ingredients and safety of the vaccine varied by community. For example, Muslim participants were very concerned about pork gelatine and the option of a pork-free vaccine [64, 66]. This again mirrors global research where the most commonly cited religiously motivated barriers to measles vaccination were animal-derived gelatins used in the vaccine production and the use of aborted human fetus tissue to make the rubella component of the MMR vaccine [72].

Additionally, for Muslim/Afro-Caribbean communities, fears about possible links to autism were also seen to be important, while these were less important for the Romani traveller community [64, 66]. This local diversity in opinion is important as a 2023 systematic review found the risk of autism to be the most commonly cited concern about the vaccine [15].

There were also issues raised about the accessibility of the vaccine services, particularly for migrants and asylum seekers [67]. A survey of support staff working with unaccompanied asylum-seeking children (UASC) living in Birmingham found that UASC often arrive with missing vaccination records, and that there often isn't enough time within appointments to give the required immunisations [65]. Additionally, a 2025 review found that many migrants living in Europe don't know how to access vaccine services and don't know what they're eligible for, including fears of costs [73]. This is a particularly unnecessary barrier since all migrants in the UK, including those with No Recourse to Public

Funds (NRPF), are eligible for free vaccinations on the NHS [74, 75]. The same review found that first-generation immigrants had low vaccination rates and high levels of unknown vaccination status, reflecting local evidence [73]. In particular, one 2020 Welsh study found that asylum-seeking children had a 3 times reduced odds of being vaccinated compared to the general population [76]. A 2019 systematic review of parental attitudes and beliefs of the measles vaccine also found that parents from Gypsy, Roma, and Traveller populations faced unique accessibility barriers due to their transient residency as well as regular discrimination and exclusion within the healthcare system [19].

Participants in the Birmingham engagement workshops also raised the issue of trust [66, 67]. Some participants reported a general mistrust of government sources, particularly if associated with the 'Home Office' and 'immigration'. A general preference was raised for information to come from healthcare professionals and local community/faith leaders. A lack of trust in the healthcare system has also been found to be a barrier to vaccination in the UK [14].

Other reasons for vaccine hesitancy include a belief that children are given too many vaccines too soon, and a low perceived benefit of vaccination [15]. A 2025 cross-sectional study in Türkiye also found parents were 2.5 times more likely to exhibit vaccine hesitancy if one of their children had already experienced adverse effects from vaccination [77].

3 Research Objectives

This project aims to answer the following research question: How can a system dynamics model be used to facilitate the development of interventions to increase measles vaccination rates and reduce negative outcomes, particularly amongst lower-uptake ethnic groups?

The research aims are as follows:

- Conduct a participatory modelling workshop with expert stakeholders to design a theoretical model for measles vaccinations and outcomes in Birmingham and Solihull, and to identify and prioritise key interventions to be modelled,
- Develop a computational model calibrated using historical vaccination, notification, and hospitalisation data,
- Integrate most highly prioritised interventions to simulate the expected impact on the modelled behavioural science factors, vaccination rates, and health outcomes.

In the long term, the aim is to develop a dashboard for health protection teams working in the system to be able to dynamically plan and test interventions using variable model assumptions. However, this is beyond the scope of the thesis.

4 Methods

4.1 Stakeholder Engagement

At the time of writing (10th March 2026), ten stakeholders from Birmingham City Council (BCC) (6), Solihull Metropolitan Council (1), BSol NHS Integrated Care Board (ICB) (2), and the UK Health Security Agency (UKHSA) (1). Of these, nine have also agreed to be part of a longer-term advisory board for this project. To help promote full participation, stakeholders will be separated into two groups. The make-up of each group will be chosen to create the best spread of background and expertise. A simplified version of the approach to stakeholder engagement used by Michail et al. will be employed for this project [78].

Before the workshop, participants will be sent an evidence pack outlining the rationale behind using system dynamics and local evidence on the systemic, sociological, and psychological barriers that prevent parents from vaccinating their children as outlined in Section 2.2.

Throughout the workshop, input from the participants will be collected on paper and sticky notes. At the end of the session, all materials will be photographed and later transcribed. For transparency, the photos and transcriptions will be made publicly available. Due to its geographical centrality and public transport connections, the workshop will be hosted at the Birmingham City Council House. The session will be broken into three parts and facilitated with the help of the BCC Governance Team.

4.1.1 Part 1 - Introduction

To ensure all participants are aware of the key issues being discussed, a high-level overview of the evidence pack will be presented and a brief overview of system dynamics. Secondly, to help everyone to conceptually understand system dynamics, there will be a short practical model-building task. To ensure all participants can participate in this task, an everyday example relating to flows of money in and out of a bank account will be used.

4.1.2 Part 2 - Model building

Stakeholders will be presented with a simplified measles model showing the main stocks and flows in the system. For each of the key flows between stocks, participants will be asked to use sticky notes to give factors that facilitate and block the flow. Once the factors are listed, participants will each be given a limited number of stickers with which they can vote on the factors for each flow that they think are dominant. While all factors are important, this will help to focus the model on those with the biggest influence on the model's behaviour.

4.1.3 Part 3 - Interventions

Participants will be asked to identify which flows they think could be improved to make the system work better. They will be asked to identify potential policies and interventions that could change this behaviour. Importantly, for each change, they will be asked to identify clearly who will be responsible for which actions and where within the system this action will occur. They will also be asked to predict in which direction this will change the flow. This information will also be collected on sticky notes.

Once all the interventions are collected, an anonymous online voting platform will then be used to allow every participant to rank each intervention by preference. This will help to focus the project on the interventions that stakeholders think are the most worth focusing on based on their feasibility and impactability.

4.2 Model Development

This will expand on the work done by Ackere and Schulz with the aim of explicitly including the role of ethnicity. Since this model is to be built collaboratively, it is not known yet what it will look like. However, the basic components included in the minimal model shown in Figure 4 will likely be included. All system dynamics modelling will be completed using Stella Architect [79]. This propriety software includes a graphical user interface that allows for models to be built visually, similar to that shown in Figure 4. This helps to engage stakeholders without technical expertise in the model building process. Ackere and Schulz and Zadeh et al. provide the parameter values used in their models, which can be used as starting points for this project [54, 56].

First, a basic version of the model outlined in the development workshop will be built. This will include all stocks and flows vectorised by broad ethnic categories (Asian, Black, Mixed, Other, and White). While this will not capture the full variation in vaccine attitudes by ethnicity, this will serve as an important first step towards building a more complete model. This will also help to avoid the impact of limited data for some of the smaller ethnic groups. Where they are known, variable values will already be included in the model. This model will be presented back to the project advisory group for validation and, where necessary, estimation of variable values.

This validated model will then be calibrated to fit historic data. The data used for calibration will include Child MMR vaccinations from the NHS Federated Data Platform [23], Primary Care registration data and hospitalisation data from the BSol ICB SQL server, and lab-confirmed measles notifications from West Midlands UKHSA (to be obtained via a freedom of information request). Some of the model variables will be left as "free" variables that can be tuned such that the model produces the expected output. As an infectious disease, measles cases occur in "outbreaks". Therefore, it will be necessary to artificially "perturb" the system by introducing a small number of infected individuals to start an outbreak.

Once a functional model of the BSol measles system has been built, the different interventions will be implemented into the model such that they can be turned on and off, if necessary, with varying

“intensity”. To use the project time as efficiently as possible, interventions will be implemented in the order of prioritisation given by stakeholders at the workshop. For each intervention, the population MMR coverage, number of measles notifications, and measles hospitalisations over time will be compared to the system without intervention.

Interventions may be preventative or reactive and will have to be evaluated separately. For preventative interventions, the system will be “perturbed” with infectious people to cause a system outbreak at fixed points in time following the intervention. Reactive interventions, however, will be activated only after the number of confirmed infections in the system has reached some threshold value.

Sensitivity analysis will also be performed. Some of the system variables will have significant uncertainty, particularly those drawn from expert opinion. For each of these values, the simulation will be re-run using plausible maximum and minimum values to assess the impact of this uncertainty on the model outputs.

All data will be exported from Stella Architect and analysed and visualised using R Studio. All code and Stella Architect models will be publicly available on GitHub³.

5 Feasibility

There are several factors that impact the feasibility of this project. Time is a significant constraint on what can be achieved. For this reason, effort has been taken to prioritise modelling efforts on aspects that are deemed to be most important by the system stakeholders (see Section 4.1). A full timeline of the project, including key deadlines, is shown in Appendix C.

One of the biggest issues is data completeness. The vaccination data coverage from the NHS federated data platform is central to this project. However, 26.5% of BSol residents in the dataset have an unknown ethnicity and, as seen from Figure 2, this cohort has the lowest MMR coverage at just 32.0%. As a first order correction, the value for these patients will be redistributed proportionally amongst the other ethnic groups. However, since some ethnic groups are more likely than others not to have their ethnicity recorded, this redistribution will be biased [80]. Similar issues are also observed in the hospitalisation and measles notifications data.

As mentioned in Section 2.1, some age-stratified disease transmission simulations consider variable epidemiological mixing between different age groups [52]. However, empirical evidence on the epidemiological mixing between ethnic groups in the UK remains limited meaning that ethnicity-specific contact matrices are currently unavailable. As a result, perfect mixing between all ethnic groups will likely have to be assumed. This will be a weakness of the model since small amount of existing evidence does point towards a degree of assortative mixing within ethnic groups [81, 82].

Another challenge relates to the novel simulation method being used. The author has completed training in building system dynamics models in Stella Architect with the BSol ICB. An academic partner at the University of Birmingham with significant experience in system dynamics and Stella Architect has also agreed to support the project. The author also has an academic background in the use of other simulation methods as well as in the use of R Studio and SQL [83–88].

Ethical approval is not required since the data used is pseudonymised and routinely collected, and the purpose of this project is for service improvement. The decision tool created by the NHS Health Research Authority also confirmed that a review by the NHS Research Ethics Committee is not required for this project. The results from this decision tool are also publicly available on the project’s GitHub repository.

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A SEIR Example

Let $S(t)$, $E(t)$, $I(t)$, $A(t)$ and $R(t)$ be the number of individuals at time t who are susceptible, exposed, infective, and recovered respectively. Since the total population is not fixed, we have $S(t) + E(t) + I(t) + A(t) + R(t) = N(t)$ for $t = \{t_1, \dots, t_k\}$.

Individuals in the Susceptible (S) compartment can become Exposed (E) through contact with Infectious (I) individuals at a constant rate β . Susceptible individuals can also die from all-cause mortality at a constant rate μ_d , become immune through vaccination at a rate ν , or revert to susceptibility from the Recovered (R) state at a rate ω . The birth rate μ_b contributes to the influx of new Susceptible individuals into the population. Exposed individuals move to the Infectious state at a rate of σ , which represents the latency period of the infection. Exposed individuals can also die from all-cause mortality at a constant rate μ_d . Infectious individuals recover and move to the Recovered state at a rate γ , or they may die due to the infection at a rate α , in addition to the all-cause mortality μ_d . Infectious individuals can develop acute poliomyelitis which leads to hospital admission at a rate η . Since hospital admission is not hospitalisation, for simplicity, we assume these individuals leave the compartment immediately in the subsequent time to the recovered compartment. Recovered individuals can lose their immunity and move back to the Susceptible state at a rate ω , or they may die from all-cause mortality μ_d . Vaccinated individuals (transitioned from Susceptible to Recovered due to vaccination at a rate ν) also follow these dynamics. The model outlines the changes in different state-spaces of the population over time t . In this model, deaths are depicted as transitions from the S, E, I, or R compartments to an external state out of the system.

These movements between states at time t can be expressed by the following differential equations:

$$\frac{dS}{dt} = \mu_b N - \frac{\beta IS}{N} - \mu_d S + \omega R - \nu S, \quad (1)$$

$$\frac{dE}{dt} = \frac{\beta IS}{N} - \sigma E - \mu_d E, \quad (2)$$

$$\frac{dI}{dt} = \sigma E - \gamma I - (\mu_d + \alpha)I - \eta I, \quad (3)$$

$$\frac{dA}{dt} = \eta I - A, \quad (4)$$

$$\frac{dR}{dt} = \gamma I - \mu_d R - \omega R + \nu S + A, \quad (5)$$

These equations can then be solved computationally to calculate the change in population in each state over time. The simplest technique is the Euler Method. This method assumes that derivatives can be approximated as constant over small time periods [89]. From this a value Y at time step $t + \Delta t$ can be approximated as

$$Y(t + \Delta t) = Y(t) + \frac{dY(t)}{dt} \Delta t. \quad (6)$$

Following this approach, Eqs 1-5 can be combined into a single matrix multiplication equation.

B Literature Search

Title	Authors	Year	Simulation Type	Citation
"Assessing measles control through vaccination and public health education: a data-driven mathematical modeling approach"	Peter et al.	2026	SEIR	[27]
"Forecasting and quantifying the role of vaccination strategies in the 2025 Texas measles outbreak: a modeling and time series analysis approach"	Chukwu	2026	SEIR + seasonality	[28]
"Modelling measles transmission dynamics and the impact of control strategies on outbreak Management"	Peter	2025	SEIR	[29]
"Modelling the effect of vaccination on the transmission dynamics of measles in China"	Liu et al.	2025	SEIR	[30]
"Assessing the Impact of Routine and Campaign Vaccination on Measles Transmission: A Modeling Study"	Sinigirira et al.	2025	SEIR	[31]
"Heuristic algorithm tool for planning mass vaccine campaigns"	Rodríguez-Pereira et al.	2025	Heuristic optimisation model	[61]
"Stochastic Milstein computing driven autoregressive exogenous neuro-architecture for chaotic nonlinear measles transmission system with impact of immunization"	Anwar et al.	2025	Stochastic model with AI emulation	[59]
"A fractional modeling approach for the transmission dynamics of measles with double-dose vaccination"	Farhan et al.	2025	SEIR	[32]
"Mathematical modelling and transmission dynamic of SEIRVQD model for epidemic diseases"	Fahimi et al.	2025	SEIR	[33]
"Deterministic and fractional-order modeling of measles dynamics with harmonic mean incidence rate and quarantine impact"	Khan et al.	2025	SEIR + memory effects	[34]
"Fitting dynamic measles models to subnational case notification data from Ethiopia: Methodological challenges and key considerations"	Sbarra et al.	2025	SEIR	[35]
"Public health interventions for developing resilience to contagious diseases: a system dynamics approach"	Zadeh et al.	2025	System Dynamics	[54]
"Long-term waning of vaccine-induced immunity to measles in England: a mathematical modelling study"	Robert et al.	2024	SEIR	[36]
"Mathematical model of measles in Turkey"	RASIT et al.	2024	SEIR	[37]
"Modelling and stability analysis of the dynamics of measles with application to Ethiopian data"	Berhe et al.	2024	SEIR	[38]
"Threshold dynamics in a periodic epidemic model with imperfect quarantine, isolation and vaccination"	Ibrahim	2024	SEIR	[39]

Table A1: Method-focused publications identified on simulations of measles transmission including vaccination as outlined in Sec 2.1 (Continued on next page).

"A compartmental approach to modeling the measles disease: a fractional order optimal control model"	Chatterjee et al.	2024	SEIR	[40]
"Analysis of the different interventions scenario for programmatic measles control in Bangladesh: A modelling study"	Kuddus et al.	2023	SEIR	[41]
"Stability and threshold dynamics in a seasonal mathematical model for measles outbreaks with double-dose vaccination"	Ibrahim et al.	2023	SEIR	[42]
"Modelling and analysis of a measles epidemic model with the constant proportional Caputo operator"	Farman et al.	2023	SEIR + memory	[43]
"Modelling, analysis, and simulation of measles disease transmission dynamics"	Alemneh et al.	2023	SEIR	[44]
"Modeling transmission dynamics of measles in Nepal and its control with monitored vaccination program"	Pokharel et al.	2022	SEIR	[45]
"A mathematical investigation of an "SVEIR" epidemic model for the measles transmission"	El Hajji et al.	2022	SEIR	[46]
"The impact of supplementary immunization activities on measles transmission dynamics and implications for measles elimination goals: A mathematical modelling study"	Shen et al.	2022	SEIR	[47]
"Mathematical analysis of a measles transmission dynamics model in Bangladesh with double dose vaccination"	Kuddus et al.	2021	SEIR	[48]
"Incorporating equity in infectious disease modeling: Case study of a distributional impact framework for measles transmission"	Menkir et al.	2021	SEIR + Income-stratification	[49]
"Insights into vaccine hesitancy from systems thinking, Rwanda"	Decouttere et al.	2021	System Dynamics	[55]
"Explaining vaccination decisions: A system dynamics model of the interaction between epidemiological and behavioural factors"	Ackere et al.	2020	System Dynamics	[56]

Table A1: (continued) Method-focused publications identified on simulations of measles transmission including vaccination as outlined in Sec 2.1.

Title	Authors	Year	Citation
"The shadow of reactogenicity: Parental hesitancy due to prior vaccine experiences"	Gschwend et al.	2026	[13]
"Religious barriers to measles vaccination"	Wombwell et al.	2015	[72]
<i>SMBC Community Engagement Learning</i>	Solihull Metropolitan Borough Council	2025	[64]
<i>UASC Health & Wellbeing Survey</i>	Kemishi	2025	[65]
"Parental vaccine hesitancy: Recent evidences support the need to implement targeted communication strategies"	Cagnotta et al.	2025	[18]
"Vaccine-preventable diseases in migrants in Europe: a systematic review"	Halliday et al.	2025	[73]
"Parental vaccine refusal, non-vaccinated children, and outbreaks of Vaccine-Preventable Diseases (VPDs) in Europe: a systematic review of aetiology and risk"	Baiocchi et al.	2025	[90]
<i>Report of Community Engagement Workshops in Birmingham: Feedback on UKHSA Published Materials on Measles</i>	Palmer et al.	2024	[66]
<i>Summary of Birmingham Measles Soft Intelligence</i>	Bissett et al.	2024	[67]
"Understanding the health system barriers and enablers to childhood MMR and HPV vaccination among disadvantaged, minority or underserved populations in middle-and high-income countries: a systematic review"	Essa-Hadad et al.	2024	[71]
"Prevalence of caregiver hesitancy for vaccinations in children and its associated factors: A systematic review and meta-analysis"	Maamor et al.	2024	[68]
"Why parents say no to having their children vaccinated against measles: a systematic review of the social determinants of parental perceptions on MMR vaccine hesitancy"	Novilla et al.	2023	[15]
"Vaccine hesitancy in migrant communities: a rapid review of latest evidence"	Tankwanchi et al.	2021	[91]
"MMR vaccine attitude and uptake research in the United Kingdom: a critical review"	Torracinta et al.	2021	[14]
"Resurgence of measles in Europe: a systematic review on parental attitudes and beliefs of measles vaccine"	Wilder-Smith et al.	2020	[19]
"Parents' and informal caregivers' views and experiences of communication about routine childhood vaccination: a synthesis of qualitative evidence"	Ames et al.	2017	[92]
"Socioeconomic differences in childhood vaccination in developed countries: a systematic review of quantitative studies"	Bocquier et al.	2017	[70]

Table A2: Publications on vaccine hesitancy.

C Timeline

